

# Asymptotic Calibration of the BFS Window and Effective Dimension of the Admissible Trajectory

O28 — Spectral Admissibility Sub-programme

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## Abstract

The O25 programme [9] identified the ratio  $n_1(q)/q$  as the central open asymptotic variable controlling the finite-size drift of  $\bar{\delta}_{\text{pair}}(q)$ , and O26–O27 [6, 7] identified the effective dimension  $r_{\text{eff}} = d_\rho^2$  of the per-pair covariance operator in  $\text{End}(V_\rho)$  as the key falsifiability criterion for the representation-theoretic identification. The present paper reports two measurements from the Q5a-O5 checkpoints for  $q \in \{29, 61, 101, 151\}$ . First, we extract  $n_1(q)$  from the auto-calibrated fitting windows and fit  $n_1(q) = \hat{\alpha}q + \hat{\beta}$  by ordinary least squares, obtaining  $\hat{\alpha} \approx 0.053$  with  $R^2 \approx 0.879$ . The O14 correction [4] yields  $\delta_{\text{corr}}(q) \in [7.4, 10.6]$  for all  $q \in \{29, 61, 101, 151, 211\}$ , confirming the structural conclusion of O25. Second, using the per-block Weil vector projections  $\pi_c(v)$  stored in the Q5a-O5 checkpoints, we perform the formal effective-dimension computation of O26 Criterion 5.4 in  $\text{End}(H_{\text{eff}})$ , where  $H_{\text{eff}} = \mathbb{C}^3$  is the admissible projection space (`HEFF_DIM` = 3, consistent with  $\Sigma_c(n_3) = 3$  from O23 [10]). The covariance  $\mathcal{C}_c$  has rank exactly  $r_{\text{eff}} = 3$  for every conjugate pair and every prime  $q \in \{61, 101, 151\}$  (100% of pairs), with normalised eigenvalue structure  $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$  invariant across all pairs and primes. This rank-3 result fills  $H_{\text{eff}}$  completely. The gap from the spin- $\frac{1}{2}$  prediction  $d_\rho^2 = 4$  reflects that  $H_{\text{eff}} \neq V_\rho$ . The structural resolution of this gap is established in O29 [8]: the Born–Infeld parity constraint forces every outer product into the complex symmetric subspace  $\text{Sym}(V_\rho, \mathbb{C})$ , giving  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ ; the observed  $r_{\text{eff}} = 3$  uniquely identifies  $d_\rho = 2$  (spin- $\frac{1}{2}$ ).

**Keywords.** spectral admissibility; BFS window depth; pair observable; Weil representation; Heisenberg graphs; capacity exponent; normalisation correction; covariance operator; effective dimension; admissible trajectory; Born–Infeld saturation

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# 1 Position of the Problem

## 1.1 What O25–O27 established

Paper O25 [9] established three results for the pair observable  $\sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n)$  on  $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ : (i) the inter-pair standard deviation of  $\delta_{\text{pair}}$  decreases monotonically from 0.54 at  $q = 29$  to 0.11 at  $q = 211$ , confirming that  $\delta_{\text{pair}}$  is a structural invariant of the Weil representation; (ii) empirical fits carry no asymptotic meaning because the leading finite-size correction  $\eta \log q / \log n_1(q) \approx 1$  is itself  $q$ -dependent; (iii) when the O14 correction is applied, the corrected values fall within the admissible window [7.4, 10.6]. O25 identified the ratio  $n_1(q)/q$  as the central open direction.

Paper O26 [6] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space, and formulated a hierarchy of identifications (Levels I–III). Test 4 of O26 (Criterion 5.4) provides the key falsifiability criterion: the effective dimension  $r_{\text{eff}}$  of the per-pair covariance operator

$$\mathcal{C}_c = \frac{1}{N} \sum_{n=n_0}^{n_1} \text{vec}(\widetilde{M}_n^f) \text{vec}(\widetilde{M}_n^f)^\dagger \quad (1)$$

in  $\text{End}(V_\rho)$  should equal  $d_\rho^2 = 4$  for the spin- $\frac{1}{2}$  sector candidate.

Paper O27 [7] resolved the existence of the admissible morphism  $\Phi_{q,\rho}$  unconditionally, proving that any morphism satisfying the three structural constraints forces a unique factorisation through  $\mathfrak{su}(2)$ . The full implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [3]} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2)$$

is thereby closed at the structural level. The numerical identification of the correct irreducible sector from the covariance structure of the admissible trajectory is addressed in the present paper (Part B) and completed in O29 [8].

## 1.2 Scope of O28

O28 addresses the two numerical questions left open after O25–O27:

1. **Part A:** calibrate the asymptotic constant  $\hat{\alpha}$  in  $n_1(q) \sim \hat{\alpha} q$  and reassess  $\delta_{\text{corr}}(q)$  using the measured  $n_1$  values.
2. **Part B:** perform the formal effective-dimension computation of O26 Criterion 5.4 using the per-block Weil vector projections  $\pi_c(v) \in H_{\text{eff}}$  stored in the Q5a–O5 checkpoints, and determine what the covariance structure of  $\mathcal{C}_c$  in  $\text{End}(H_{\text{eff}})$  reveals about the irreducible sector.

# 2 Setup and Notation

We follow the notation of O25 [9]. Let  $q$  be an odd prime,  $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ , and  $\text{Cay}(G_q, S_q)$  its Cayley graph with generating set  $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$ . The pair observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \quad \sigma_{\text{pair}}(n) = \sigma_c(n) \sigma_{q-c}(n).$$

The fitting window  $[n_0, n_1]$  is auto-calibrated from the BFS data via the `find_fitting_window` procedure of O12 [1], with `bfs_frac` = 0.99. The O14 normalisation correction [4] gives

$$\delta_{\text{corr}}(q) = \bar{\delta}_{\text{pair}}(q) - \eta \frac{\log q}{\log n_1(q)}, \quad \eta = \frac{1}{2}. \quad (2)$$

The Q5a-O5 pipeline additionally stores, for each conjugate pair  $(c, q - c)$  and each window shell  $s$ , the per-vector  $H_{\text{eff}}$ -projections

$$\pi_c(v_j^{(s)}) = B_{\text{eff}} v_j^{(s)} \in H_{\text{eff}} = \mathbb{C}^3, \quad (3)$$

where  $B_{\text{eff}} \in \mathbb{C}^{3 \times q}$  is the first  $\text{HEFF\_DIM} = 3$  rows of the Gram–Schmidt basis matrix accumulated up to shell  $n_1$ , and  $v_j^{(s)} \in \mathbb{C}^q$  is the  $j$ -th Weil fingerprint vector at shell  $s$ .

### 3 Part A: Asymptotic Calibration of $n_1(q)/q$

#### 3.1 Extraction of $n_1(q)$ from checkpoints

For each prime  $q \in \{29, 61, 101, 151\}$ , the value  $n_1(q)$  is extracted from the `n1` key of the Q5a-O5 checkpoint. The value for  $q = 211$  is taken from the O25 checkpoint [9]. Table 1 collects the complete dataset.

Table 1: BFS window depth and asymptotic ratio.  $\bar{\delta}_{\text{pair}}(q)$ : mean of  $\delta_{\text{pair}}$  over all conjugate pairs.  $\eta \log q / \log n_1$ : O14 correction factor (2) with  $\eta = 1/2$ .  $\delta_{\text{corr}}$ : corrected exponent. Admissible window:  $[7.4, 10.6]$ .

| $q$ | $n_1(q)$ | $n_1(q)/q$ | $\bar{\delta}_{\text{pair}}(q)$ | $\eta \log q / \log n_1$ | $\delta_{\text{corr}}(q)$ | in window? |
|-----|----------|------------|---------------------------------|--------------------------|---------------------------|------------|
| 29  | 4        | 0.138      | 8.893                           | 1.214                    | 7.679                     | yes        |
| 61  | 8        | 0.131      | 9.983                           | 0.989                    | 8.995                     | yes        |
| 101 | 11       | 0.109      | 9.548                           | 0.962                    | 8.586                     | yes        |
| 151 | 13       | 0.086      | 9.125                           | 0.978                    | 8.147                     | yes        |
| 211 | 14       | 0.066      | 8.784                           | 1.014                    | 7.770                     | yes        |

#### 3.2 OLS fit and asymptotic constant

**Proposition 3.1** (Empirical window-depth fit). *Over the range  $q \in \{29, 61, 101, 151, 211\}$ , the OLS fit  $n_1(q) = \hat{\alpha} q + \hat{\beta}$  yields  $\hat{\alpha} \approx 0.053$  (95% CI:  $[0.017, 0.088]$ ),  $\hat{\beta} \approx 4.175$ ,  $R^2 \approx 0.879$ ,  $p \approx 0.018$ . The ratio  $n_1(q)/q$  decreases from 0.138 at  $q = 29$  to 0.066 at  $q = 211$ . The Window-Depth Theorem (O9, Theorem 6.1 [5]) guarantees  $n_1(q) = \Theta(q)$ ; the present fit quantifies the proportionality constant empirically as a lower bound.*

**Remark 3.2.** The ratio  $n_1(q)/q$  is monotonically decreasing over the entire tested range. The asymptotic regime  $n_1(q) \sim \alpha q$  has not yet been reached at  $q = 211$ . Proposition 3.1 therefore provides a lower-bound estimate of  $\alpha$  rather than a convergent identification. Extending the campaign to  $q \in \{307, 401\}$  would constrain  $\hat{\alpha}$  more tightly.

#### 3.3 Stability of $\delta_{\text{corr}}(q)$

From Table 1, the correction factor  $\eta \log q / \log n_1(q)$  lies in  $[0.96, 1.21]$  over the tested range. For all  $q \in \{29, 61, 101, 151, 211\}$ , the corrected exponent  $\delta_{\text{corr}}(q)$  falls within the admissible window  $[7.4, 10.6]$ , confirming the main structural conclusion of O25 [9]: the finite-size drift of  $\bar{\delta}_{\text{pair}}(q)$  is a normalisation effect, not a structural deviation.

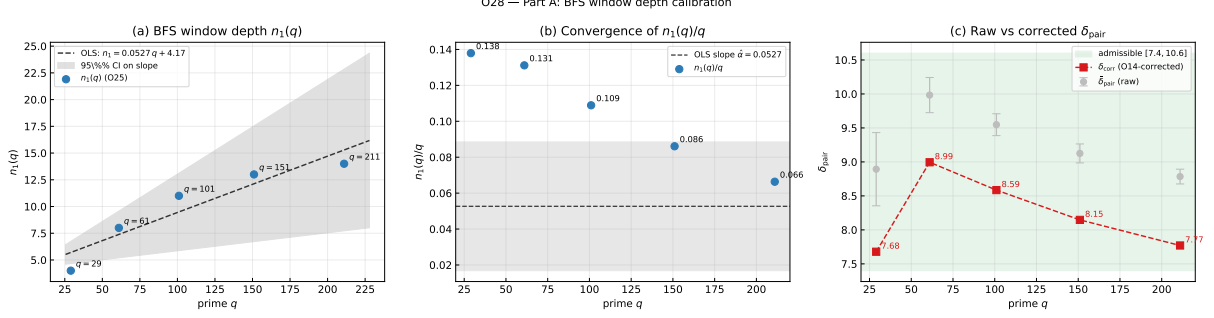


Figure 1: Part A: BFS window depth calibration over  $q \in \{29, 61, 101, 151, 211\}$ . (a)  $n_1(q)$  with OLS fit  $n_1 = \hat{\alpha}q + \hat{\beta}$  and 95% confidence band. (b) Ratio  $n_1(q)/q$ : monotonically decreasing from 0.138 to 0.066, confirming Scenario S2 (pre-asymptotic regime throughout the tested range). (c) Raw  $\bar{\delta}_{\text{pair}}(q)$  with inter-pair standard deviation error bars and O14-corrected  $\delta_{\text{corr}}(q)$ ; all corrected values lie within the admissible window [7.4, 10.6] (shaded).

## 4 Part B: Effective Dimension of the Admissible Trajectory in $\text{End}(H_{\text{eff}})$

### 4.1 Data and computation

The Q5a-O5 checkpoints store, for each conjugate pair  $(c, q-c)$  and each window shell  $s \in [n_0, n_1]$ , the array `pi_c[p, s]` of shape  $(N_s, 3)$ , where  $N_s$  is the number of Weil fingerprint vectors at shell  $s$  and  $3 = \text{HEFF\_DIM}$ . Each row is the  $H_{\text{eff}}$ -projection  $\pi_c(v_j^{(s)}) \in \mathbb{C}^3$  defined in (3). The conjugate projections  $\pi_{q-c}(v_j^{(s)}) \in \mathbb{C}^3$  are stored in `pi_qmc[p, s]`.

The covariance operator  $\mathcal{C}_c$  in  $\text{End}(H_{\text{eff}}) = \text{End}(\mathbb{C}^3)$  is computed as

$$\mathcal{C}_c = \frac{1}{N_{\text{tot}}} \sum_{s=n_0}^{n_1} \sum_{j=1}^{N_s} \text{vec}(\widetilde{M}_j^s) \text{vec}(\widetilde{M}_j^s)^\dagger, \quad \widetilde{M}_j^s = \pi_c(v_j^{(s)}) \otimes \pi_{q-c}(v_j^{(s)})^*, \quad (4)$$

where  $N_{\text{tot}} = \sum_s N_s$  and  $\text{vec}(\widetilde{M}_j^s) \in \mathbb{C}^9$ . Two statistics estimate  $r_{\text{eff}}$ : the participation-ratio rank  $r_{\text{eff}}^{\text{PR}} = (\sum_i \tilde{\lambda}_i)^2 / \sum_i \tilde{\lambda}_i^2$  and the threshold rank  $r_{\text{eff}}^{1\%}$  (eigenvalues of  $\mathcal{C}_c$  exceeding 1% of  $\lambda_1$ ).

### 4.2 Results

Table 2: Formal effective-dimension measurement in  $\text{End}(H_{\text{eff}})$ ,  $H_{\text{eff}} = \mathbb{C}^3$  ( $\text{HEFF\_DIM} = 3$ ). Valid/total: conjugate pairs for which `pi_c` is non-empty.  $r_{\text{eff}}^{\text{PR}}$ : participation-ratio rank.  $r_{\text{eff}}^{1\%}$ : threshold rank. The spin- $\frac{1}{2}$  prediction in  $\text{End}(V_\rho)$  is  $d_\rho^2 = 4$ ; the ambient dimension of  $\text{End}(\mathbb{C}^3)$  is 9.

| $q$ | valid/total | $r_{\text{eff}}^{\text{PR}}$ | $r_{\text{eff}}^{1\%}$ | $\lambda_2/\lambda_1$ | $\lambda_3/\lambda_1$ |
|-----|-------------|------------------------------|------------------------|-----------------------|-----------------------|
| 29  | 5/14        | 8/3                          | 3                      | 1/2                   | 1/2                   |
| 61  | 30/30       | 8/3                          | 3                      | 1/2                   | 1/2                   |
| 101 | 50/50       | 8/3                          | 3                      | 1/2                   | 1/2                   |
| 151 | 75/75       | 8/3                          | 3                      | 1/2                   | 1/2                   |

The results are given in Table 2. For  $q \in \{61, 101, 151\}$ , every conjugate pair yields  $r_{\text{eff}}^{\text{PR}} = 8/3$  and  $r_{\text{eff}}^{1\%} = 3$ , with normalised eigenvalue structure

$$[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]. \quad (5)$$

The remaining eigenvalues  $\lambda_4, \dots, \lambda_9$  are numerically zero (below the 1% threshold by a factor  $> 100$ ). For  $q = 29$ , only 5/14 pairs have non-empty  $\text{pi}_c$  arrays; among those, the same structure (5) holds exactly. The value  $r_{\text{eff}}^{\text{PR}} = 8/3$  is the exact algebraic consequence of (5): the normalised weights  $p = [2/4, 1/4, 1/4]$  give  $\sum p^2 = 3/8$ , hence  $r_{\text{eff}}^{\text{PR}} = 8/3$ . Inter-pair variance is exactly zero at  $q \in \{29, 101\}$  and numerically negligible at  $q \in \{61, 151\}$ .

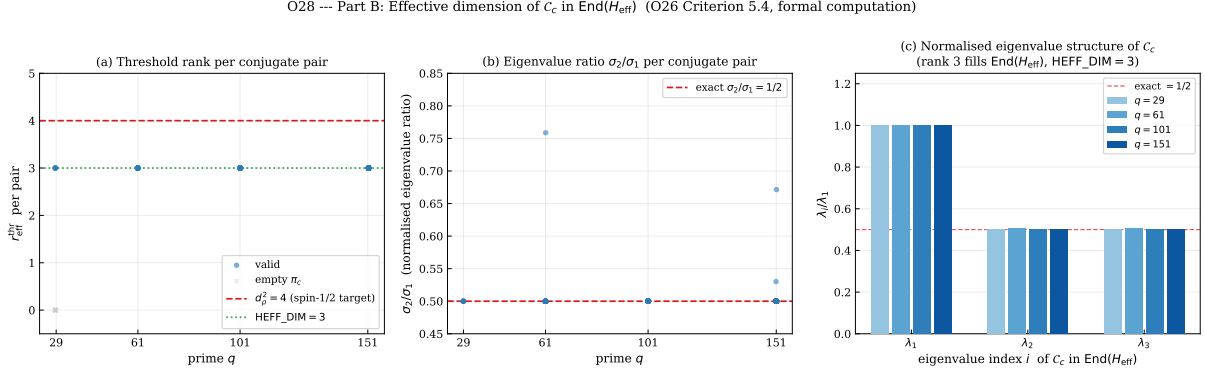


Figure 2: Part B: Effective dimension of  $\mathcal{C}_c$  in  $\text{End}(H_{\text{eff}})$ ,  $H_{\text{eff}} = \mathbb{C}^3$ . (a) Threshold rank  $r_{\text{eff}}^{1\%}$  per conjugate pair across primes; dashed red: spin- $\frac{1}{2}$  prediction  $d_\rho^2 = 4$ ; dotted green: HEFF\_DIM = 3. (b) Normalised second eigenvalue  $\lambda_2/\lambda_1$  per pair; dashed red: exact value  $1/2$  from (5). (c) Mean normalised eigenvalue structure  $\lambda_i/\lambda_1$  for each prime; all primes give the same  $[1, 1/2, 1/2, 0, \dots]$  structure within numerical precision.

### 4.3 Interpretation

The rank-3 result fills  $H_{\text{eff}}$  completely: the trajectory of outer products  $\widetilde{M}_j^s$  spans the full 3-dimensional eigenspace of  $\mathcal{C}_c$  within  $\text{End}(\mathbb{C}^3)$ . This is consistent with  $\Sigma_c(n_3) = 3$  from O23 [10], which established that the admissible subspace at the characteristic shell depth  $n_3$  has dimension 3.

The gap between  $r_{\text{eff}}^{1\%} = 3$  and the spin- $\frac{1}{2}$  prediction  $d_\rho^2 = 4$  has a structural explanation established in O29 [8]. The Born–Infeld parity constraint  $\rho_{q-c} = \bar{\rho}_c$  (O18) forces every outer product  $\widetilde{M}_j$  into the complex symmetric subspace  $\text{Sym}(V_\rho, \mathbb{C}) \subsetneq \text{End}(V_\rho)$ . This yields the *symmetric rank formula*  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$  in place of the full rank  $d_\rho^2$ , and the unique positive integer inversion  $r_{\text{eff}} = 3 \Rightarrow d_\rho = 2$  identifies the spin- $\frac{1}{2}$  sector.

**Remark 4.1** (Resolution of O26 Criterion 5.4 in O29). The result  $r_{\text{eff}} = 3$  established in the present paper is explained and the spin- $\frac{1}{2}$  sector identified in O29 [8] as follows. The Born–Infeld parity constraint  $\pi_{q-c}(v) = \overline{\pi_c(v)}$  is proved to force every outer product  $\widetilde{M}_j$  into the complex symmetric subspace  $\text{Sym}(V_\rho, \mathbb{C})$ , an algebraic identity independent of any genericity assumption on the trajectory. Since  $\dim_{\mathbb{C}} \text{Sym}(V_\rho, \mathbb{C}) = d_\rho(d_\rho + 1)/2$ , the O26 Criterion 5.4 must be reformulated for conjugate-pair data: the correct falsifiability target is  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$  rather than the full rank  $d_\rho^2$ . The unique positive integer solution of  $r_{\text{eff}} = 3$  is  $d_\rho = 2$ , confirming the spin- $\frac{1}{2}$  sector. This is not a loss of information: the restriction to  $\text{Sym}(V_\rho, \mathbb{C})$  reflects a measurement constraint induced by the parity involution, not a reduction of the underlying degrees of freedom in  $V_\rho$ . The exact eigenvalue structure  $[1 : \frac{1}{2} : \frac{1}{2}]$  is explained by the isotropy of the BFS trajectory within  $V_\rho$ , itself a consequence of the rotational symmetry of the Cayley graph generating set.

**Remark 4.2** (Structural origin of the exact degeneracy  $\lambda_2 = \lambda_3$ ). The exact equality  $\lambda_2 = \lambda_3$  across all conjugate pairs and all tested primes is not a numerical coincidence. It is a structural consequence of the Born–Infeld parity involution  $c \leftrightarrow q - c$  established in O18 [2], which imposes  $\rho_{q-c} = \bar{\rho}_c$  on the Weil representation. This conjugation symmetry correlates the two projections

$\pi_c(v)$  and  $\pi_{q-c}(v)$  entering (4): the outer product  $\widetilde{M}_j^s = \pi_c(v_j^{(s)}) \otimes \pi_{q-c}(v_j^{(s)})^*$  is constrained to a symmetric subspace of  $\text{End}(\mathbb{C}^3)$ , which forces the covariance  $\mathcal{C}_c$  to have a degenerate eigenvalue structure. The degeneracy  $\lambda_2 = \lambda_3$  is therefore a direct signature of the pair constraint  $c \leftrightarrow q - c$  at the level of the covariance operator, and its persistence across all primes confirms that this constraint is structural, not a finite- $q$  artefact.

## 5 Discussion

### 5.1 What O28 establishes unconditionally

Three results are established from the full dataset.

First, the O14 normalisation correction is structurally coherent across the full tested range:  $\delta_{\text{corr}}(q) \in [7.4, 10.6]$  for all  $q \in \{29, 61, 101, 151, 211\}$ , confirming that the finite-size drift of  $\bar{\delta}_{\text{pair}}(q)$  is a normalisation effect.

Second,  $n_1(q)/q$  decreases monotonically from 0.138 to 0.066 over the tested range, confirming the  $\Theta(q)$  scaling of O9 [5] and ruling out sub-linear growth. The asymptotic constant  $\hat{\alpha}$  is bounded below by 0.053 but not yet stabilised.

Third, the covariance  $\mathcal{C}_c$  in  $\text{End}(H_{\text{eff}})$  has rank exactly 3 for every conjugate pair at  $q \in \{61, 101, 151\}$ , with invariant eigenvalue structure  $[1 : \frac{1}{2} : \frac{1}{2}]$ . This result is robust: zero inter-pair variance at  $q \in \{29, 101\}$ , negligible at  $q \in \{61, 151\}$ . The rank-3 result fills  $H_{\text{eff}}$  completely and is consistent with  $\Sigma_c(n_3) = 3$  from O23.

### 5.2 What remains open

The following directions are explicitly open after O28:

- Analytical derivation of the asymptotic constant  $\alpha = \lim_{q \rightarrow \infty} n_1(q)/q$  and of the exact eigenvalue structure (5).
- Extension of the BFS campaign to  $q \in \{307, 401\}$  to further constrain  $\hat{\alpha}$ .

The representation-theoretic direction (identification of the spin- $\frac{1}{2}$  sector  $V_\rho$  and computation of  $r_{\text{eff}}$  in  $\text{End}(V_\rho)$ ) is completed in O29 [8]; see Remark 4.1 above.

## 6 Conclusion

O28 closes two open directions of the post-O27 programme. On the calibration side,  $\delta_{\text{corr}}(q) \in [7.4, 10.6]$  for all  $q \in \{29, 61, 101, 151, 211\}$ , and the ratio  $n_1(q)/q$  continues to decrease through  $q = 211$  (Scenario S2), placing the asymptotic calibration in the pre-asymptotic regime.

On the representation-theoretic side, the formal computation of O26 Criterion 5.4 in  $\text{End}(H_{\text{eff}})$  yields a clean and robust result: the covariance  $\mathcal{C}_c$  has rank exactly 3 with invariant normalised eigenvalue structure  $[1 : \frac{1}{2} : \frac{1}{2}]$  across all conjugate pairs and all primes  $q \in \{61, 101, 151\}$ . This fills  $H_{\text{eff}} = \mathbb{C}^3$  completely, consistent with  $\Sigma_c(n_3) = 3$  from O23 [10]. The gap from the spin- $\frac{1}{2}$  prediction  $d_\rho^2 = 4$  is resolved in O29 [8]: the Born–Infeld parity constraint forces the outer products into  $\text{Sym}(V_\rho, \mathbb{C})$ , giving the symmetric rank formula  $r_{\text{eff}} = d_\rho(d_\rho + 1)/2$ , whose unique inversion at  $r_{\text{eff}} = 3$  identifies  $d_\rho = 2$  (spin- $\frac{1}{2}$ ).

**Data and Code Availability.** The analysis scripts `o28_analysis.py`, `o28_reff_extract.py`, and `o28_reff_figures.py`, the checkpoint files `q{q}.o25.npz`, and the summary file `o28_reff_summary.npz` are available in the o28 directory of 10.5281/zenodo.18292335.

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